

Boammani Aser Lompo

MSc

RESEARCH & WORK EXPERIENCE



About me

ML researcher (MILA/ÉTS) working on **mechanistic interpretability** and the safety of model editing, on a strong theoretical base (ENS, Agrégation, IMO). End-to-end research delivery: ICML and NeurIPS-workshop papers, an implicit-bias theory preprint, and open-source benchmarks and codebases.

Webpage

<https://aser97.github.io/Blog/>

Areas of specialization

Interpretability • AI Safety • LLM • Learning Theory • Maths

Interests

- Photography & Content Creation
- **Competitive Chess (2000 Elo)**



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2023–now

Graduate Researcher, MILA / ÉTS Montréal

RESEARCHER • Montréal 📍

Mechanistic interpretability of arithmetic circuits in LLMs (GPT-J, Llama-3-8B): circuit identification via Edge Pruning validated against Tracr ground truth, SAE feature tracking, and inference-time substitution of a learned sub-computation by an exact external module. Central question: are feature *directions* more task-exclusive than the *nodes* carrying them — can an edit be certified not to damage tasks sharing the same weights. Also: implicit bias of GD under spurious correlations (theory); multimodal reasoning benchmarks.

2021

Internship, Department of Statistics, CREST, ENSAI

INTERN • Rennes 📍

Research on unimodal bandits for ranking systems. Keywords: Recommender Systems, Multi-armed Bandits, Learning to Rank. Supervisor: Romaric Gaudel

2020

Internship in a Research Laboratory in Cryptology, INRIA

INTERN • Paris 📍

Measuring privacy of cryptographic systems and studying the properties of optimal-privacy systems. Supervisor: Catuscia Palamidessi

DEGREES

2025 MSc Machine Learning Research

MILA / ETS MONTRÉAL
• Ecole Technologie Supérieure (Montreal) 🏛️

2023 MSc Mathematics, Vision & Learning

MVA • Ecole Normale Supérieure Paris-Saclay 🏛️

2021 MSc Fundamental Mathematics; Agrégation (French national competitive exam)

AGRÉGATION • Ecole Normale Supérieure (Rennes) 🏛️

TECH STACK

Languages: Python • C++ • JavaScript • Bash
Frameworks: PyTorch • TransformerLens • SAELens • Transformers • JAX • vLLM • LoRA/PEFT
Tools: Docker • Weights & Biases • Paperspace • Git • MCP
Focus Areas: Interpretability • AI Safety • Foundation Models • Learning Theory

PORTFOLIO

N2P •Mechanistic interp.: circuit ID + SAE tracking, **GPT-J / Llama-3-8B**
•Direction-level exclusivity as an **edit-safety criterion**
•Github Writeup

Safety Guardrail •Fine-tuned **Qwen2.5-7B** + LoRA
•9-signal youth distress taxonomy
•**F1 = 0.85** Github

Visual-TableQA •**2.5k** complex table images, **9k** QA pairs
•Cost-efficient pipeline (**<\$100**), **+17%** fine-tuning gain
•Github

Number Encoding •LESA + Xval encoder upgrade for LLMs
•**+12%** on numerical reasoning Github
More on my webpage.

Awards & Olympiads

- 2014** **Honourable Mention**, International Mathematical Olympiad (**IMO**), Cape Town
- 2015** **Gold medal**, Panafrican Mathematics Olympiad (**PAMO**), Abuja
- 2015** **Winner**, National Mathematics Contest, Burkina Faso
- 2024** Best Oral Presentation, **Acfas 2024** (91^e Congrès)

Grants

- 2024** R-SADC scholarship (**23,000\$**)
- 2018–22** French Government Excellence Grant (**50,000\$**)

Languages

- | | |
|----------------|-------------|
| English | Fluent (C2) |
| French | Native (C2) |

Publications

- 2026** B.A. Lompo, P. Kenfack. *Implicit Bias of Gradient Descent under Feature-Mediated Spurious Correlations*. **Preprint, under submission**. [Page] [GitHub]
 - 2025** B.A. Lompo, M. Haraoui. *Modality-Swap Distillation: Rendering Textual Reasoning into Visual Supervision*. **Preprint**.
 - 2025** B.A. Lompo, M. Haraoui. *Visual-TableQA: Open-Domain Benchmark for Reasoning over Table Images*. **NeurIPS 2025** Workshop. [PDF]
 - 2025** B.A. Lompo, T-D. Le, P. Jouvet, R. Noumeir. *Multi-objective Representation for Numbers in Clinical Narratives*. **under submission**. [PDF]
 - 2021** C-S. Gauthier, R. Gaudel, E. Fromont, B.A. Lompo. *Parametric Graph for Unimodal Ranking Bandit*. **ICML 2021**. [PDF]
- Reviewer:** ICCV 2023 • CVPR 2026 • ICLR 2026

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